# Participant Questionnaire

**Development of Haptic Wearable Devices for Morse Code Communication**

*Please do not write your name or any personal details anywhere on this form. Your answers are completely anonymous. You may leave any question blank if you prefer not to answer it.*

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## Part 1. Background

1.1 Before today, how familiar were you with Morse code?

☐ Not at all ☑ I had heard of it ☐ I knew a few characters ☐ I knew it well

1.2 Do you regularly wear a device that vibrates (for example a smartwatch or fitness band)?

☐ Yes ☑ No

## Part 2. Feeling the vibrations

*1 = strongly disagree 2 = disagree 3 = neutral 4 = agree 5 = strongly agree*

| Statement | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| I could easily feel the vibrations on my wrist. | ☐ | ☐ | ☐ | ☐ | ☑ |
| I could clearly tell the difference between short and long vibrations. | ☐ | ☐ | ☐ | ☑ | ☐ |
| I could tell where one character ended and the next began. | ☐ | ☑ | ☐ | ☐ | ☐ |
| The vibration patterns were easy to follow at the speed they were sent. | ☐ | ☐ | ☐ | ☑ | ☐ |

2.5 The strength of the vibration was:

☐ Too weak ☐ Slightly weak ☑ About right ☐ Slightly strong ☐ Too strong

## Part 3. Tapping messages

*1 = strongly disagree 2 = disagree 3 = neutral 4 = agree 5 = strongly agree*

| Statement | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| It was easy to tap out dots and dashes on the device. | ☐ | ☐ | ☐ | ☐ | ☑ |
| I felt confident that my taps were being registered correctly. | ☐ | ☐ | ☐ | ☐ | ☑ |
| I could tap at a speed that felt natural to me. | ☐ | ☐ | ☐ | ☐ | ☑ |

## Part 4. Comfort and wearability

*1 = strongly disagree 2 = disagree 3 = neutral 4 = agree 5 = strongly agree*

| Statement | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| The device was comfortable to wear. | ☐ | ☐ | ☐ | ☑ | ☐ |
| I would be willing to wear the device for a longer period of time. | ☐ | ☐ | ☑ | ☐ | ☐ |

## Part 5. Learning Morse code with the companion display

*This section is about the screen (phone or computer) that displayed the message while you felt it as vibration.*

*1 = strongly disagree  2 = disagree  3 = neutral  4 = agree  5 = strongly agree*

| Statement | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Seeing the message on the screen helped me understand what I was feeling. | ☐ | ☐ | ☐ | ☐ | ☑ |
| The screen helped me connect particular vibration patterns to particular characters. | ☐ | ☐ | ☐ | ☐ | ☑ |
| By the end of the session, I could recognise some characters without looking at the screen. | ☐ | ☐ | ☐ | ☐ | ☑ |
| I think this setup (vibration together with on-screen text) would help me learn Morse code over time. | ☐ | ☐ | ☐ | ☐ | ☑ |
| I would prefer learning Morse code this way over learning it from a book or app alone. | ☐ | ☐ | ☑ | ☐ | ☐ |
| I found myself relying on the screen rather than paying attention to the vibration. | ☐ | ☐ | ☐ | ☑ | ☐ |

5.7 How useful did you find the on-screen display overall?

☐ Not useful  ☐ Slightly useful  ☑ Moderately useful  ☐ Very useful  ☐ Essential

5.8 If you used this device regularly, how long do you think you would need the screen before you could rely on vibration alone?

☐ I could already  ☑ A few sessions  ☐ Several weeks  ☐ I would always need it  ☐ Not sure

## Part 6. Overall

*1 = strongly disagree  2 = disagree  3 = neutral  4 = agree  5 = strongly agree*

| Statement | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Overall, the device was easy to use. | ☐ | ☐ | ☐ | ☐ | ☑ |
| I would be interested in using a device like this. | ☑ | ☐ | ☐ | ☐ | ☐ |

6.3 What did you find most difficult about using the device?

I struggled to differentiate the space between dots and dashes vs. space between letters.

6.4 What would you change or improve?

- Seeing the morse code chart and live keying.
- Smaller

6.5 Any other comments?

For learning, an app alone would be just as useful.
I could learn morse code in a few sessions but it would take longer to become proficient.
If I need to communicate via morse code I would likely use the device.